

What Remains Useful After Strong Frontier Prompting? Complementary Local Models for Ultra-Low-Resource Tangut Title Translation

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Abstract

Tangut short-text translation combines two difficulties: the data are extremely scarce, and the targets are often compact historical titles rather than ordinary modern sentences. In our repository snapshot, only 491 real Tangut-Chinese pairs are available, which makes the central question sharper than in ordinary low-resource MT. Once a strong frontier Chinese LLM is already equipped with dictionary glosses and a title-oriented prompt, what remains worth learning locally? We answer this with a layered case study: first establish a deliberately strong frontier prompt-only upper bound, then compare synthetic local supervised fine-tuning (SFT) recipes and filtered direct preference optimization (DPO), and finally test whether frontier and local candidates can be combined through several open rerankers and a closed catalog-aware adjudicator. The paper is intentionally framed as a controlled benchmark and workflow comparison rather than as a scholar-in-the-loop deployment study: the goal here is to narrow the most reliable model and reranking families under matched references, while expert-facing utility validation is reserved for follow-up work once the candidate workflow has stabilized. The pattern is consistent. Frontier prompting is the strongest single-model strategy. Within the open/local regime, structured multitask supervision is the right base, and DPO helps only after aggressive preference-gap filtering. Local training is still useful not because it replaces frontier prompting, but because it contributes conservative alternative titles that improve downstream selection: simple open reranking already helps, whereas generic consensus maximization can overshoot and damage title quality. A supplementary 200-item canonicalized oracle-bone stress test over EVOBC IDs shows the same directional pattern under a deliberately minimal transfer setup: dictionary-grounded prompting is the strongest single oracle model, but a simple open 2-way hybrid with MT-SFT still lifts exact match from 40/200 to 44/200, reaches 47.81 chrF++, and reduces contamination to 0.5%. The broader lesson is to frame ultra-low-resource historical translation as workflow design under reference-aware evaluation, since lexical coverage and perplexity can mis-rank the gold reference itself.

1 Introduction

Historical translation at the ultra-low-resource end is difficult for a different reason than ordinary low-resource MT. The bottleneck is not only data scarcity, but also output shape: many targets are compact titles, glosses, or catalog strings rather than fluent modern sentences. Ancient-language surveys, ancient Chinese translation bakeoffs, and historical-document modernization studies provide partial precedents, but they usually target broader sentence translation or orthographic updating rather than compact cross-script title reconstruction [1, 2, 3]. Tangut is a sharper case still. In the repository studied here, only 491 real Tangut-Chinese pairs are available, and many targets are short title-like strings rather than ordinary sentences. A system can therefore sound plausible while still missing the real scholarly target, namely the correct historical title in the correct form.

This changes the paper’s central question. Once a strong frontier Chinese LLM is already given dictionary glosses and a title-oriented prompt, the issue is no longer whether a local model can simply beat it head-to-head. The more useful question is what remains worth learning locally, and whether local hypotheses can still improve a controlled workflow under matched references.

We therefore organize the paper around three research questions:

1. **RQ1:** How far can strong frontier prompting go as a single-model upper bound?
2. **RQ2:** If local training still matters, what kind of supervision signal helps most?
3. **RQ3:** Do frontier and local systems fail differently enough to yield a reproducible workflow gain?

We answer these questions by comparing local dictionary-grounded prompting, a deliberately strong frontier prompt-only baseline, synthetic pseudo-parallel SFT, preference optimization on top of SFT, and inference-time selection over frontier and local candidates [5, 4, 11]. The design is layered on purpose: first establish the strongest single-model upper bound, then identify the best open/local recipe, and only then ask whether complementarity can be turned into a workflow gain. To keep that final claim honest, we also include simple open reranking controls such as plurality vote, pairwise chrF-MBR, and a local Qwen reranker over the same candidate pool.

The resulting story is simple. Frontier prompting is the strongest single-model strategy. Within the open/local regime, structured multitask supervision is the right base, and DPO helps only after aggressive preference-gap filtering. The main positive result is not local replacement of the frontier system, but recovery of complementary conservative hypotheses through reranking. The open rerankers already provide a reproducible gain, while the stronger closed adjudicator is used only as an upper bound on recoverable complementarity. A supplementary oracle-bone stress test then checks portability without moving the paper’s center: dictionary-grounded prompting remains the strongest single oracle model, but a simple open 2-way hybrid with MT-SFT still improves over either component alone.

This framing is deliberate. We do not position the paper as a scholar-in-the-loop utility study. A small expert panel would answer a different question from the one studied here and would still be too small to support strong claims about expert consensus. We therefore first narrow the model and workflow story under matched references, then reserve expert-facing validation for follow-up external evaluation.

The paper makes three contributions:

1. a research-question-driven case study that separates three claims often blurred together in low-resource work: the strongest single-model upper bound, the best open/local trainable recipe, and the extent to which complementary local hypotheses can be recovered in a workflow under only 491 real pairs;
2. a positive local-method result showing that structured multitask supervision is the right base, that DPO helps only after aggressive preference-gap filtering, and that local models matter because they contribute complementary conservative hypotheses rather than because they dominate the frontier model head-to-head;
3. a reference-aware evaluation and reranking analysis showing why short historical titles must not be judged primarily by generic reference-free metrics, and how different open rerankers behave on the same frontier/local candidate pool.

Tangut remains the only benchmark here with matched real short-title translations and the only setting where the full frontier/local workflow claim is made.

2 Related Work

Ancient and historical language processing. Ancient-language NLP differs from mainstream NLP not only because data are scarce, but also because scripts, editorial conventions, and annotation practices are unstable across corpora [1]. Recent ancient-Chinese translation bakeoffs and historical-document modernization studies operate under similar sparsity, though they typically target sentence translation or orthographic updating rather than cross-script title reconstruction [2, 3]. Our Tangut setting is narrower still: only 491 real pairs are available, and the evaluation set is dominated by compact title-like strings.

Low-resource MT, synthetic supervision, and lexical grounding. Low-resource MT often relies on synthetic parallel data, multilingual transfer, lexicon integration, or explicit decoding constraints [4, 6, 8, 5, 7]. Tangut overlaps with this literature but is harsher in two ways: the source script is largely outside standard MT pretraining, and the outputs are compact catalog titles rather than ordinary sentences. We therefore focus on synthetic supervision and lexicon-grounded structure rather than on broad multilingual transfer.

Dictionary-structured prompting and error-aware prompting. Recent LLM translation work suggests that lexical resources are most useful when presented as structured intermediate evidence rather than as flat prompt context. Chain-of-Dictionary prompting and Learning-From-Mistakes prompting both follow this pattern in low-resource settings [9, 10]. Our stronger frontier baseline is adjacent to this line, but the present paper asks a different question: whether alignment structure should be absorbed into trainable local targets, and whether those local hypotheses remain useful after a strong frontier prompt-only baseline is already in place.

Preference optimization. Direct Preference Optimization (DPO) is now a standard offline alignment objective, and recent work has adapted direct preference or quality optimization to machine translation while also studying robustness under noisy feedback [11, 14, 15, 12, 13]. Our results place this literature in a harsher regime: supervision is tiny, outputs are title-like, and the chosen/rejected signal is built from partially misaligned automatic metrics. In this setting, DPO helps only after base-model choice and pair filtering are controlled aggressively.

Evaluation for non-standard translation outputs. chrF++ is widely used in machine translation because character-level overlap is often more stable than token-level overlap under orthographic variation [16]. Learned MT metrics and QE systems such as COMET and OpenKiwi go beyond surface overlap, but recent work also shows that they can have domain-specific failure modes and induce metric interference [17, 20, 18, 19]. Our findings sharpen that caution in a historical short-text regime: lexical coverage and perplexity can penalize the gold reference itself, so reference-aware evidence must remain primary.

3 Task Setting

The paper studies Tangut short-text translation under two coupled constraints. First, supervision is tiny: the repository contains only 491 real pairs, split into 400 training examples, 41 development examples, and 50 test examples. Second, the outputs are short and often title-like. Success therefore depends on precise lexical choice and compact formatting rather than generic sentence fluency.

This makes the task narrow but meaningful. We ask how to recover short historical titles under only 400 real training examples, not how to solve open-domain Tangut translation. To train local models at all, we use synthetic data as structured proxies: each SFT branch adds 50,000 synthetic Tangut-Chinese pairs built from monolingual historical Chinese and dictionary knowledge, then combines them with the real training set upsampled by a factor of 10. These synthetic examples are not claims about faithful Tangut reconstruction; they are judged only by held-out performance on real Tangut titles.

We also audit data hygiene conservatively. There is zero exact source, target, or full-pair overlap across train/dev/test, zero exact overlap between the test references and the final targets of the 50k synthetic multitask corpus, and zero exact overlap between the frontier few-shot exemplars and the test references. Approximate n -gram overlap between synthetic multitask targets and the test references is also low, so the synthetic set is not acting as a near-duplicate cache.

Evaluation is the second bottleneck. We therefore treat chrF++, exact match, contamination, suffix preservation, and the reference-aware judge as primary evidence, while lexical coverage, perplexity, and the legacy reference-free judge remain diagnostic. chrF++ is a reasonable overlap anchor for this setting because character-level scoring is relatively stable under orthographic variation [16], whereas perplexity and lexical coverage can mis-rank even the gold reference in this title-heavy regime [17, 20, 18, 19].

4 Methods

We design the comparison around the research questions rather than around every implementation variant. The repository contains the full ablation matrix; the main text keeps only the method families needed to answer whether strong prompting is already enough, what local signal helps most, and whether frontier/local complementarity can be turned into a workflow.

4.1 Prompt-Only Upper Bounds

The prompt-only family contains no weight updates. **baseline1** prompts the local model directly on the Tangut string, **baseline2** adds dictionary glosses, and **baseline2_1_cot** further decomposes generation into alignment, literal drafting, and rewriting. This family is closest to lexicon-integrated or lexically constrained MT, except that the lexicon is injected through prompting rather than decoder constraints [5, 7].

We then add a deliberately strong frontier baseline based on **deepseek/deepseek-v3.2** via OpenRouter with dictionary grounding and four fixed development-set exemplars spanning canonical, bodhisattva-practice, dharani-style, and transliteration-heavy titles. The prompt enforces compact title form, forbids explanatory paraphrase, and emits only the final title string. Decoding uses temperature 0.0 and a maximum of 512 output tokens, so the frontier baseline is deterministic and there is no sampling-seed variance to report. This baseline is intentionally strong: if frontier prompting already solves the task, the local-training story should shrink accordingly.

To separate prompt wording from model scale, we also rerun the same frontier few-shot title prompt on local Qwen2.5-7B. A constrained-decoding control keeps that prompt fixed and adds local constrained beam search with at most one short content constraint plus one title-suffix constraint, retaining content phrases only when the 1–2 character phrase also appears in the 441 real non-test Tangut titles.

4.2 Local Training Recipes

We build 50k pseudo-Tangut/Chinese pairs from monolingual historical Chinese by mapping Chinese characters to Tangut characters through dictionary explanations. The four SFT branches differ only in how dictionary misses are handled and whether the target includes alignment structure:

- mixed-script keeps some Chinese characters in the source;
- UNK replaces misses with [UNK];
- semantic projection replaces misses through embedding lookup;
- multitask keeps the UNK input but changes the target to `<dict_match>{...}</dict_match><literal>{...}</literal> {final title}`.

Each synthetic set is combined with the 400 real training examples upsampled by a factor of 10. In the main narrative, UNK-SFT isolates the effect of input purity and multitask SFT tests whether explicit alignment structure helps further. The mixed and semantic variants remain in the repository but do not change the conclusions.

4.3 Preference Alignment on Local Bases

The DPO family asks a narrower question: once a strong local base exists, can reward-based alignment improve it without damaging title form? We apply DPO to an SFT base using reward-defined chosen/rejected pairs [11]. For each training prompt we sample five outputs and score them with

$$r(y) = \text{lex}(y) - 0.01 \log \text{ppl}(y),$$

where $\text{lex}(y)$ is the repository’s lexical-coverage score. The $\text{ppl}(y)$ term is computed by a local Qwen2.5-0.5B causal LM, and we use only the raw $\log \text{ppl}$ value with coefficient 0.01, without additional normalization. The reward therefore favors dictionary consistency with only a weak fluency tie-breaker, which is why we treat it cautiously later.

The legacy pipeline constructs preferences from the best pre-multitask SFT base, yielding the `final_v2` checkpoint as a negative control. The follow-up DPO runs instead start from multitask SFT. After auditing the preference data, we keep only pairs whose reward gap exceeds either 0.2 or 0.4 and train both standard sigmoid DPO and a robust+weighted variant [12, 13]. This turns the DPO family into a controlled test of base-model choice, pair filtering, and loss robustness.

4.4 Hybrid Adjudication

The final method family operationalizes the paper’s main positive claim. We build an inference-time adjudicator over the frontier DeepSeek baseline, UNK-SFT, and MT-DPO g0.4 sig. The closed adjudicator uses Azure-hosted GPT-5.4 and may either copy one candidate or make a minimal repair from candidate fragments. Its prompt treats the task as historical catalog-title reconstruction rather than free paraphrase, and the repository stores the exact templates and per-example JSON traces.

For reproducibility, we also add a deterministic open selector over the same 3-way pool. It uses title-suffix priors, candidate-length agreement, contamination penalties, conservative guards against truncation or expansion, and a small frontier prior. We further compare three simpler open controls over the identical pool: plurality vote, pairwise chrF-MBR, and a local Qwen2.5-7B reranker that can only copy one candidate verbatim. These baselines help separate task-aware conservative selection from generic agreement maximization.

4.5 Supplementary EVOBC Portability Stress Test

Reviewer questions naturally target portability across historical scripts rather than Tangut alone, so we add a second-script stress test built around EVOBC oracle-bone IDs. Each external EVOBC ID is compiled into a reversible private-use surrogate codepoint so that the Tangut dictionary and training pipeline can be reused without changing the single-character source contract. The transferable part of the method is therefore the symbolic interface and the local-training stack, not the Tangut codepoints themselves.

To move beyond a pure infrastructure port, we derive a real held-out probe from OBIMD sentence-level reading sequences by canonicalizing them into the EVOBC inventory through shared modern-character labels and sampling length-balanced 50/50/200 train/dev/test splits. This yields a deliberately conservative decipherment stress test rather than a second short-title benchmark. The oracle setup reuses the symbolic interface and synthetic local-training stack, but not the Tangut-specific reranker or reference-aware judge.

One evaluation detail matters. The oracle multitask SFT checkpoint still emits `<dict_match>`, then `<literal>`, and then a freer Chinese explanation, mirroring the Tangut pipeline. The canonicalized oracle probe instead targets only the compact normalized decipherment string. We therefore report the oracle SFT model primarily through the `<literal>` field and keep the raw assistant tail as a control for output-format mismatch.

Because the main Tangut claim is workflow-level complementarity rather than local single-model dominance, we also evaluate a deliberately simple open 2-way oracle selector over the dictionary-grounded prompt and the MT-SFT `<literal>` output. If this minimal selector can already recover extra correct items, then the complementary local-repair behavior has transferred in the sense relevant to the paper.

4.6 Implementation Details

All learned models start from Qwen2.5-7B-Instruct with LoRA adapters. SFT uses three epochs, learning rate 2×10^{-4} , batch size 4, gradient accumulation 4, rank-64 LoRA, and maximum sequence length 512. DPO uses two epochs, learning rate 5×10^{-5} , batch size 1, gradient accumulation 8, $\beta = 0.1$, rank-32 LoRA, cosine decay, and the same sequence length; the robust variants add label smoothing and WPO-style weighting. The reference-aware judge is a supplementary evaluator, not a training signal. It uses Azure-hosted GPT-5.4 to score reference agreement, source faithfulness, title-style fitness, and overall quality after seeing the Tangut source, gold reference, and candidate simultaneously.

5 Main Results

Table 1 reports only representative systems. The empirical story is short: a strong frontier baseline already answers much of the single-model question, local learning helps only when it stays close to title structure, and the main reproducible gain comes from selecting among complementary candidates rather than asking one local model to replace the frontier system.

5.1 RQ1: Is Strong Frontier Prompting Already Enough as a Single-Model Upper Bound?

Yes. The DeepSeek frontier baseline is far stronger than the local prompt baselines, reaching 28.54 chrF++, 12/50 exact matches, and 2.80 overall under the reference-aware judge. Prompt-only is

Method	chrF++ $\uparrow$	Exact Match $\uparrow$	Ref.-Aware Overall $\uparrow$
<i>Prompt-only</i>			
Dict prompt	6.13	0/50	1.50
<i>Frontier DeepSeek</i>	28.54	12/50	2.80
<i>Local learned</i>			
UNK-SFT	22.11	3/50	1.82
MT-SFT	25.99	5/50	1.98
Legacy DPO	19.39	2/50	1.68
<i>MT-DPO g0.4 sig</i>	30.01	5/50	2.22
<i>Hybrid workflow</i>			
<i>Open selector (3-way, open)</i>	32.62	12/50	2.90
Catalog adjudicator (3-way, closed)	34.87	14/50	3.02
Human ref.	100.00	50/50	5.00

Table 1: Representative systems for the first three research questions. Frontier DeepSeek is the strongest single model, MT-DPO g0.4 sig is the strongest local learned model, and the guarded open 3-way selector is the main reproducible workflow. Additional open controls appear in Table 3; the closed adjudicator is reported only as headroom.

Local Prompt Control	chrF++ $\uparrow$	Exact Match $\uparrow$	Contam. $\downarrow$
Legacy local dict+CoT prompt	7.59	1/50	5/50
Frontier-style few-shot prompt	12.18	1/50	3/50
+ lexical constrained beam search	12.35	1/50	4/50

Table 2: Reviewer-facing local prompt/control comparison. The second row reuses the exact DeepSeek-style few-shot title prompt on Qwen2.5-7B; the third adds conservative lexical constrained beam search. Prompt alignment helps relative to the legacy local CoT prompt, but lexical constraints add only a marginal chrF++ gain.

therefore not a straw baseline here. A strong frontier system is already hard to beat, so every later claim has to explain what local training still contributes after that bar is in place. The frontier configuration is also fully enumerated now: **deepseek/deepseek-v3.2** via OpenRouter, temperature 0.0, maximum 512 output tokens, and four fixed development-set exemplars. Because decoding is deterministic, sampling seeds are not relevant. We do not claim a full paid prompt-variance sweep, but the stored pilot variants in the repository support the chosen prompt direction: on the shared 5-item pilot slice, the final few-shot title-only prompt reaches 2/5 exact matches and 67.68 chrF++, versus 0/5 and 21.99 chrF++ for the strict title-only prompt.

The new local prompt controls answer two reviewer questions directly. Reusing the exact frontier-style few-shot prompt on local Qwen2.5-7B lifts chrF++ from 7.59 for the legacy local CoT prompt to 12.18 and reduces contamination from 5/50 to 3/50, but exact match stays at 1/50. Adding constrained beam search over one filtered content phrase plus one suffix constraint moves chrF++ only marginally to 12.35, again with 1/50 exact matches and slightly worse contamination at 4/50. The frontier advantage is therefore not just better wording wrapped around the same local model, and the missing constrained-decoding baseline does not explain away the single-model gap.

Workflow	Pool	Exact Match	chrF++	Overall
<i>Frontier DeepSeek</i>	single	12/50	28.54	2.80
<i>Plurality vote (open)</i>	F+UNK+DPO	13/50	30.06	2.94
<i>Local Qwen reranker (open)</i>	F+UNK+DPO	12/50	30.85	2.82
<i>Guarded selector (open)</i>	F+UNK+DPO	12/50	32.62	2.90
<i>Pairwise chrF-MBR (open)</i>	F+UNK+DPO	11/50	34.76	2.64
3-way catalog adjudicator (closed)	F+UNK+DPO	14/50	34.87	3.02
Best-of-3 oracle	F+UNK+DPO	14/50	41.85	3.12

Table 3: Workflow comparison with standard open reranking controls. “F” = frontier DeepSeek, “UNK” = UNK-SFT, and “DPO” = MT-DPO g0.4 sig. The guarded selector is the strongest clean open workflow; the closed adjudicator and oracle are upper bounds only.

5.2 RQ2: If Local Training Still Matters, What Signal Helps Most?

The answer is not that more local training is always better. What helps is supervision that explicitly teaches the model how a short historical title is assembled. UNK-SFT already improves over prompt-only local decoding by enforcing pure Tangut inputs, and multitask SFT improves again, making structured supervision the right base. DPO is not uniformly helpful: the legacy DPO checkpoint trained on the old UNK-selected base drops to 19.39 chrF++ and 1.68 overall, so reward-based alignment is clearly not a blanket win. Once DPO is moved onto the multitask base and filtered aggressively, however, it becomes useful: MT-DPO g0.4 sig is the strongest local model at 30.01 chrF++ and 2.22 overall. The gap-0.4 robustWPO control reaches the same overall score but lower chrF++, so the main gain is still better explained by cleaner labels and the stronger multitask base than by a more elaborate loss alone.

5.3 RQ3: Can Complementary Failures Be Turned into a Stronger Workflow?

Yes, and this is where the paper’s main positive result appears. The frontier model remains the best single predictor, but the best local models are still useful because they fail differently and often more conservatively on short titles. Table 3 includes not only our guarded selector, but also standard open reranking controls over the same 3-way frontier/UNK/DPO pool. Even a trivial plurality vote already improves the frontier baseline to 30.06 chrF++, 13/50 exact matches, and 2.94 overall, which shows that complementarity is visible without any learned scorer. A local Qwen reranker reaches 30.85 chrF++ and 2.82 overall. Pairwise chrF-MBR pushes overlap higher to 34.76 chrF++, yet it over-switches and hurts judged quality, dropping to 2.64 overall. The guarded heuristic is therefore not the only useful open reranker, but it is the strongest clean overlap-oriented one, reaching 32.62 chrF++ and 2.90 overall with zero contamination. The 3-way catalog-aware adjudicator still goes further to 34.87 chrF++, 3.02 overall, and 14/50 exact matches, but we treat that result only as a closed upper bound on recoverable complementarity.

The workflow result should therefore be read in two layers: open reranking over frontier and local candidates already helps reproducibly, and the closed adjudicator shows that the same pool still contains additional recoverable complementarity. Pool ablations and sensitivity checks in the repository do not change this ordering.

5.4 Supplementary Oracle Portability Probe

Table 4 reports a deliberately conservative second-script stress test: a 200-item held-out canonicalized OBIMD→EVOBC probe. It evaluates text-side decipherment over real OBIMD reading

Method	Setting	Exact Match	chrF++	Contam.
Qwen2.5-7B zero-shot	single model	0/200	0.44	27.5%
Qwen2.5-7B + dict prompt	strongest single model	40/200	46.20	3.5%
MT-SFT (<literal>)	single local model	29/200	40.56	5.0%
<i>Simple 2-way selector (open)</i>	dict + MT-SFT	44/200	47.81	0.5%
Best-of-2 oracle	dict + MT-SFT	48/200	51.47	2.0%

Table 4: Supplementary 200-item canonicalized OBIMD→EVOBC portability probe. The multi-task model is read through its <literal> field, and a simple open 2-way selector lifts the dictionary-grounded prompt from 40/200 to 44/200 exact matches while reducing contamination from 3.5% to 0.5%.

sequences after canonicalization into the EVOBC inventory and does not reuse the Tangut-specific reranker or the reference-aware judge. Zero-shot Qwen largely refuses the surrogate input, but dictionary grounding immediately makes prompt-only transfer usable at 40/200 exact matches and 46.20 chrF++.

The key portability signal is therefore not local single-model dominance, but survival of complementary repair behavior under a harsher interface. Reading the oracle multitask checkpoint through its <literal> field yields 29/200 exact matches and 40.56 chrF++, and a simple open 2-way selector already lifts the dictionary-grounded prompt from 40/200 to 44/200 exact matches while reducing contamination from 3.5% to 0.5%. This supports the same workflow-level claim as the main Tangut benchmark.

6 Analysis

6.1 Why the workflow gain is conservative rather than verbose

Table 5 summarizes the output-side diagnostics. Here “contamination” is a deterministic binary flag rather than a tuned threshold: an output is contaminated iff it contains any Latin alphabetic character, angle-bracket markup, an explicit [UNK] token, or one of the small prompt-artifact lexemes observed in the legacy runs. “Suffix preservation” is computed on references containing at least one canonical title-suffix character; a prediction is counted as correct when it preserves all such reference-side suffix characters. The table caption gives the exact suffix list.

The frontier baseline is already a clean single model: 12/50 exact matches, 0 contamination, 24/27 suffix preservation, and a near-reference length ratio of 0.95. Within the local family, multi-task SFT is the cleanest pure SFT system, and strict gap-0.4 DPO regularizes the outputs further. The 3-way open selector stays clean as well, while the 3-way catalog hybrid is cleaner still with 14/50 exact matches, 26/27 suffix preservation, 0 contamination, and the same 0.95 length ratio as the frontier baseline. The gain therefore does not come from verbose rewriting; it comes from better conservative title reconstruction.

Table 6 then operationalizes the companion error-taxonomy question. We group non-exact outputs into contamination, truncation, over-expansion, title-form slips (suffix drop or numeral mismatch), and residual content drift. This matters because the failure profiles are not interchangeable. Legacy DPO is the only representative system with material artifact leakage (9/50), while frontier prompting is already clean and mostly fails through content drift plus a single severe truncation. MT-SFT and MT-DPO remain title-like more often than the legacy DPO checkpoint, but they still expose more suffix and numeral slips than the frontier baseline. The guarded open selector inherits the frontier cleanliness, removes the frontier truncation case, and keeps the re-

Method	Exact Match	Contam. Rate	Suffix Preserve	Len. Ratio
<i>Prompt-only</i>				
Dict prompt	0/50	0.00	3/27	1.83
<i>Frontier DeepSeek</i>	12/50	0.00	24/27	0.95
<i>Local learned</i>				
UNK-SFT	3/50	0.06	17/27	1.05
MT-SFT	5/50	0.02	19/27	1.10
Legacy DPO	2/50	0.18	18/27	1.88
<i>MT-DPO g0.4 sig</i>	5/50	0.04	22/27	1.12
<i>Hybrid workflow</i>				
<i>Open selector (3-way, open)</i>	12/50	0.00	23/27	0.98
Catalog adjudicator (3-way, closed)	14/50	0.00	26/27	0.95
Human ref.	50/50	0.00	27/27	1.00

Table 5: Reference-aligned diagnostics. “Contam.” denotes mixed-script or prompt-artifact contamination; “Suffix Preserve” is computed on references containing at least one title-suffix character in {經論記疏頌儀義傳錄贊序字品}.

Method	Exact	Contam.	Trunc.	Over-exp.	Suffix/Num.	Drift
<i>Frontier DeepSeek</i>	12	0	1	1	9	27
MT-SFT	5	1	0	5	12	27
<i>MT-DPO g0.4 sig</i>	5	2	0	2	7	34
<i>Open selector (3-way, open)</i>	12	0	0	1	8	29
Catalog adjudicator (3-way, closed)	14	0	0	0	7	29
Legacy DPO	2	9	0	2	8	29

Table 6: Deterministic failure taxonomy on the 50-item Tangut test set. “Contam.” is a binary artifact flag with no tuned threshold; “Suffix/Num.” groups title-like cases that miss a reference-side suffix or mismatch the numeral string.

maining errors in the same conservative title-reconstruction regime; the closed catalog adjudicator reduces the length-error bucket further, but we still report it only as headroom because its decision rule is not fully open.

6.2 Why three candidates are enough

The oracle analysis mattered because it showed that frontier and local candidates were not near-duplicates. The 3-way deterministic selector over frontier/UNK-SFT/MT-DPO already changes 12/50 cases without any closed model or text synthesis, while the closed catalog adjudicator still selects directly from the pool on 48/50 items and performs only 2 minimal repairs. The exact-match union of the three candidate sources is also 14/50, so the closed adjudicator closes the exact oracle gap without relying on free rewriting.

The open reranking controls then clarify what kind of complementarity is present. Plurality vote changes only 2/50 frontier outputs yet still reaches 13 exact matches and 2.94 overall, so some local repairs are strong enough to surface through exact agreement alone. A local Qwen reranker remains clean but only reaches 2.82 overall, while pairwise chrF-MBR over-switches, raising contamination and lowering judged quality despite higher overlap. The full 3-way selector matters because it combines multiple conservative local alternatives rather than relying on one lucky exact-match

Method	Ref. Agr.	Src. Faith.	Title Style	Overall
<i>Prompt-only</i>				
Dict prompt	1.50	1.52	2.10	1.50
<i>Frontier DeepSeek</i>	2.78	2.82	4.48	2.80
<i>Local learned</i>				
UNK-SFT	1.78	1.84	4.36	1.82
MT-SFT	1.92	1.98	4.18	1.98
Legacy DPO	1.64	1.72	3.80	1.68
<i>MT-DPO g0.4 sig</i>	2.12	2.26	4.60	2.22
<i>Hybrid workflow</i>				
<i>Open selector (3-way, open)</i>	2.82	2.88	4.50	2.90
Catalog adjudicator (3-way, closed)	2.94	3.00	4.72	3.02
Human ref.	5.00	5.00	5.00	5.00

Table 7: Representative reference-aware judge scores on a 1–5 scale. The judge sees the Tangut input, the gold reference, and the candidate output simultaneously, so this table is closer to the true task than the legacy reference-free judge.

override. Qualitatively, it restores 五部經 instead of the frontier normalization 五類經 and fixes the frontier truncation 番 to 番言金剛王乘根.

6.3 Why reference-aware evaluation is necessary

The legacy LLM judge in the repository is reference-free, so we reran evaluation with a stricter judge that sees the Tangut input, the gold reference, and the candidate simultaneously.

Table 7 changes the ranking in the direction the task actually demands. Among single models, the frontier baseline remains the clear winner at 2.80 overall; within the local learned family, MT-DPO g0.4 sig is strongest at 2.22. Among workflows, the open selector reaches 2.90 overall and the closed 3-way catalog adjudicator 3.02, with especially large gains on non-suffix and very short titles. A local Qwen2.5-7B judge preserves the same frontier/open/hybrid/human ordering (4.20, 4.20, 4.24, 4.90), reducing concern that the ranking is purely a GPT-family artifact.

The **human_reference** run is an important anchor. The gold reference receives perfect chrF++ and perfect reference-aware judge scores, but only moderate lexical coverage and catastrophic perplexity. Lexical coverage therefore measures overlap with dictionary-derived semantics rather than agreement with the gold title, while perplexity penalizes compact title style and historical transliterations. These metrics remain useful diagnostics, but they should not drive model selection in this task.

6.4 Why filtered DPO changes the local story

Table 9 summarizes the robustness checks around the local and workflow claims.

Table 8 makes the small-test-set issue explicit rather than implicit. On Tangut, the exact-match and chrF++ intervals are wide enough that the paper should avoid triumphalist claims about small workflow deltas. This is why the Tangut result is now phrased conservatively: open reranking is directionally helpful and reproducible, but only the closed adjudicator is clearly separated from the frontier baseline on paired chrF++ bootstrap. The paired bootstrap comparisons mirror the same story: filtered DPO improves the local base, plurality vote and the guarded selector remain directionally positive but boundary-level, and pairwise chrF-MBR is a useful negative control because

Suite	Method	Exact Match (Wilson 95%)	chrF++ (bootstrap 95%)
Tangut	<i>Frontier DeepSeek</i>	12/50 [14.3, 37.4]%	28.54 [20.87, 37.60]
Tangut	<i>Open selector (3-way, open)</i>	12/50 [14.3, 37.4]%	32.62 [25.08, 41.19]
Tangut	Catalog adjudicator (3-way, closed)	14/50 [17.5, 41.7]%	34.87 [26.69, 44.40]
Tangut	MT-DPO g0.4 sig	5/50 [4.4, 21.4]%	30.01 [22.09, 38.68]
EVOBC	Qwen2.5-7B + dict prompt	40/200 [15.1, 26.1]%	46.20 [42.10, 50.34]
EVOBC	<i>Simple 2-way selector (open)</i>	44/200 [16.8, 28.2]%	47.81 [43.38, 52.18]
EVOBC	Best-of-2 oracle	48/200 [18.6, 30.4]%	51.47 [47.43, 55.75]

Table 8: Uncertainty estimates for the main Tangut and EVOBC headline systems. chrF++ intervals come from bootstrap resampling and exact-match intervals from Wilson 95% confidence intervals.

Comparison ($B - A$)	Δ chrF++	Δ Overall	Δ Exact
Vote – Frontier	+1.52 [0.00, 5.14]	+0.14 [−0.04, 0.36]	+0.02 [0.00, 0.06]
Local Qwen – Frontier	+2.31 [0.11, 5.48]	+0.02 [−0.10, 0.14]	0.00 [0.00, 0.00]
Guarded open – Frontier	+4.08 [−2.28, 10.64]	+0.10 [−0.20, 0.40]	0.00 [−0.08, 0.08]
chrF-MBR – Frontier	+6.22 [−2.60, 14.60]	−0.16 [−0.50, 0.18]	−0.02 [−0.08, 0.04]
Closed 3-way – Frontier	+6.33 [2.10, 11.67]	+0.22 [0.00, 0.48]	+0.04 [0.00, 0.10]
MT-DPO g0.4 sig – MT-SFT	+4.02 [0.84, 7.80]	+0.24 [0.00, 0.46]	0.00 [0.00, 0.00]

Table 9: Paired bootstrap comparisons with 5,000 resamples. “Overall” denotes mean reference-aware judge score and “Exact” denotes exact-match rate.

its overlap gain does not survive judged-quality checks. The overlap audit addresses a separate concern: there is zero exact source, target, or full-pair overlap across train/dev/test, zero exact overlap between test references and the 50k synthetic multitask targets, and zero exact overlap between the frontier few-shot exemplars and the test references.

The DPO audit then explains why the local story changes once the labels are cleaned. Across all 3,867 preference pairs, the reward-chosen candidate is closer to a simple gold proxy only 71.3% of the time, but this rises to 81.5% for gaps ≥ 0.40 , so the high-gap filter is isolating cleaner labels rather than acting as a cosmetic hyperparameter. The reward-design ablation sharpens the same conclusion: removing the perplexity tie-breaker changes essentially nothing, whereas adding multitask `<literal>` overlap yields much cleaner but much sparser labels. We therefore interpret the DPO result narrowly: strict filtering rescues a noisy lexical reward well enough to make DPO competitive, but a stronger structure-aware reward remains future work. Guarded-selector sensitivity runs show the same kind of stability on the workflow side, so we keep the default selector as the overlap-oriented open operating point.

7 Discussion and Limitations

This paper is a question-driven case study of what changes once a strong frontier model is already available under extreme scarcity. With only 491 real pairs, the issue is not whether one local model can beat that frontier system outright, but what local signal remains worth learning. In this repository, the answer is structural conservatism: alignment-scaffolded local training produces title candidates that fail differently from frontier prompting and are useful mainly as complements. The most reproducible positive claim is therefore workflow-level, not single-model replacement.

The scope nevertheless remains narrow. The current data support Tangut short-text translation,

Audit item	Value
Total preference pairs	3867
Duplicate chosen/rejected pairs	3
Non-finite reward entries	3
Mean reward gap	0.2396
Chosen closer to gold proxy	71.3%
Chosen closer on gap (0.05, 0.10]	62.3%
Chosen closer on gap (0.20, 0.40]	75.4%
Chosen closer on gap ≥ 0.40	81.5%
Pairs kept with gap ≥ 0.20	1996
Pairs kept with gap ≥ 0.30	1010
Pairs kept with gap ≥ 0.40	498

Table 10: Lightweight audit of the current DPO pair construction pipeline. Higher reward gaps correlate with cleaner preference labels under a simple gold-proxy check.

Reward design	Flips vs current	Keep ≥ 0.2	Good @ 0.2	Good @ 0.4
Current: lex $-0.01 \log \text{ppl}$	0	1996	77.0%	81.5%
Lex only	0	1994	77.3%	81.8%
<i>Literal chrF++ only</i>	499	166	94.6%	88.0%
<i>0.5 lex + 0.5 literal chrF++</i>	21	832	85.1%	86.8%
Oracle final-target chrF++	1078	869	99.9%	100.0%

Table 11: Diagnostic reward-design ablation on the existing 3,867 DPO pairs. “Good” denotes the fraction of retained pairs whose higher-reward candidate is closer to the synthetic final target under a simple sequence-overlap proxy.

especially title-like strings, not open-domain sentence translation. The strongest adjudicator and primary reference-aware judge are still closed models, and the new oracle-bone result is still a small canonicalized decipherment probe rather than a matched short-title benchmark. The next steps follow directly from these limits: distill the closed adjudication signal into stronger open reranking, expand the oracle side from canonicalized OBIMD→EVOBC probes to richer matched oracle short-text references, and only then run targeted expert evaluation.

8 Conclusion

Tangut short-text translation is difficult here because the data are tiny and the desired outputs are compact historical titles that generic MT objectives, metrics, and rerankers model poorly. After strong dictionary-grounded frontier prompting is already in place, local training remains useful only when it supplies conservative complementary titles that open rerankers can exploit. The new controls sharpen that lesson: simple vote already helps, guarded selection is the strongest clean open workflow, and generic pairwise MBR overshoots. The supplementary oracle-bone result now reinforces the same workflow-level point in a second script. On a small canonicalized OBIMD→EVOBC stress test, dictionary-grounded prompting remains the best single model, but a simple open 2-way hybrid with MT-SFT still lifts exact match from 40/200 to 44/200 and reduces contamination to 0.5%. The title-shaped workflow components should nevertheless still be treated as Tangut-specific until they are validated on richer oracle references. This paper stops at controlled comparison rather than claiming scholar-facing deployment readiness. Ultra-low-resource historical translation

is better framed here as workflow design under reference-aware evaluation than as a single-model race or a premature usability study. The immediate next step is to distill the closed adjudication signal into stronger open reranking, expand the oracle side from canonicalized probes to matched oracle short-text references, and only then run expert-facing validation as a follow-up external check on the narrowed workflow.

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